width=!,height=!,

1. a) In the cylindrical coordinate system, $\hat{\alpha_r}$'s direction is a function of the angle ϕ, as measured counterclockwise from the positive x-axis, and restricted to the range $0 \leq \phi < 2\pi$. In summary, we may assume an angle ϕ between $\hat{\alpha_x}$ and $\hat{\alpha_r}$.

   Next, we may assume an angle of 90° between $\hat{\alpha_r}$ and $\hat{\alpha_\phi}$, since all basis vectors in the cylindrical coordinate system are orthogonal, and so define the angle between $\hat{\alpha_x}$ and $\hat{\alpha_\phi}$ as $\phi + 90°$. By the definition of the dot product, $\hat{\alpha_\phi} \cdot \hat{\alpha_x} = |\hat{\alpha_\phi}||\hat{\alpha_x}|cos(\theta_{\hat{\alpha_\phi}\hat{\alpha_r}})$, where $\theta_{\hat{\alpha_\phi}\hat{\alpha_r}}$ is the angle between the two vectors. Since the magnitude of both vectors $|\hat{\alpha_r}| = |\hat{\alpha_\phi}| = 1$,

$$\hat{\alpha_\phi} \cdot \hat{\alpha_r} = cos(\theta_{\hat{\alpha_\phi}\hat{\alpha_r}}) = cos(\phi + 90°) = -sin(\phi).$$

   We can double check this result by using the coordinate system transforms given in Table 3-2 of the textbook, where $\hat{\alpha_x} = \hat{\alpha_r}cos(\phi) - \hat{\alpha_\phi}sin(\phi)$; and so, keeping in mind that the dot product is distributive,

$$\hat{\alpha_\phi} \cdot \hat{\alpha_x} = \hat{\alpha_\phi} \cdot (\hat{\alpha_r}cos(\phi) - \hat{\alpha_\phi}sin(\phi)) = \hat{\alpha_\phi} \cdot \hat{\alpha_r}cos(\phi) - \hat{\alpha_\phi} \cdot \hat{\alpha_\phi}sin(\phi).$$

   Note that multiplying either unit vector by a scalar does not affect the angular relationship between it and another vector. Since $\hat{\alpha_\phi}$ and $\hat{\alpha_r}$ are orthogonal, their dot product is equal to 0, while the dot product between the identical vectors $\hat{\alpha_\phi}$ is equal to 1. Thus, the dot product simplifies to

$$\hat{\alpha_\phi} \cdot \hat{\alpha_r} = -sin(\phi).$$

   b) $\hat{\alpha_R}$'s direction is a function of both the zenith angle θ and the azimuth angle ϕ, similar to how $\hat{\alpha}_r$ is determined relative to ϕ in the cylindrical coordinate system. In this case, we can locate the unit vector $\hat{\alpha_R}$ at an angle θ from the positive z-axis and an angle ϕ from the positive x-axis in the rectangular system.

   Transforming $\hat{\alpha_R}$ into the rectangular coordinate system,

$$\hat{\alpha_R} = \hat{\alpha_x}sin(\theta)cos(\phi) + \hat{\alpha_y}sin(\theta)sin(\phi) + \hat{\alpha_z}cos(\theta).$$

   Once again applying the distributive property of the dot product and reducing orthogonal vectors to zero, as in question 1a, we are left with

$$\hat{\alpha_R} \cdot \hat{\alpha_y} = sin(\theta)sin(\phi).$$

   c) First, defining any cross product $\vec{A} \times \vec{B} = |A||B|sin(\theta_{AB})\hat{n}$, where $\hat{n}$ is the unit normal vector, whose direction is found using the right hand rule, we assume as before that, since $\hat{\alpha_r}$ and $\hat{\alpha_R}$ are unit vectors and have a magnitude of 1, their cross product simplifies to $\hat{\alpha_r} \times \hat{\alpha_R} = sin(\theta_{AB})\hat{n}$.

The cross product $\vec{A} \times \vec{B}$ may also be expressed in the form of a determinant (in cylindrical coordinates), as

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{\alpha_r} & \hat{\alpha_\phi} & \hat{\alpha_z} \\ A_r & A_\phi & A_z \\ B_r & B_\phi & B_z \end{vmatrix}$$

Transforming the unit vector of $\hat{\alpha_R} = \hat{\alpha_r} sin(\theta) + \hat{\alpha_z} cos(\theta)$, we may express the cross product between the two given unit vectors as

$$\hat{\alpha_r} \times \hat{\alpha_r} = \hat{\alpha_r} \times \hat{\alpha_r} sin(\theta) + \hat{\alpha_r} \times \hat{\alpha_z} cos(\theta).$$

Assuming that the cross product between the identical vectors $\hat{\alpha_r}$ equals zero, irrespective of either vector's scaling, we may then express the cross product in determinant form as

$$\hat{\alpha_r} \times \hat{\alpha_z} cos(\theta) = \begin{vmatrix} \hat{\alpha_r} & \hat{\alpha_\phi} & \hat{\alpha_z} \\ A_r & A_\phi & A_z \\ B_r cos(\theta) & B_\phi cos(\theta) & B_z cos(\theta) \end{vmatrix}.$$

Since both quantities are unit vectors, the entire determinant simplifies to $-\hat{\alpha_\phi} cos(\theta)$. Therefore,

$$\hat{\alpha_r} \times \hat{\alpha_R} = -\hat{\alpha_\phi} cos(\theta).$$

This concurs with the use of the right-hand rule, which indicates that any cross-product between the two unit vectors should be orthogonal to the ϕ-z plane, depending on the order in which the cross product is taken.

d) In this case, we may transform $\hat{\alpha_\theta}$ into its cylindrical components as $\hat{\alpha_\theta} = \hat{\alpha_r} cos(\theta) - \hat{\alpha_z} sin(\theta)$, which expands the given cross product to

$$\hat{\alpha_r} cos\theta - \hat{\alpha_z} sin(\theta) \times \hat{\alpha_z}.$$

Making use of the anti-commutative and distributive properties of the cross product, this may be further simplified to

$$-[\hat{\alpha_z} \times \hat{\alpha_r} cos(\theta) + \hat{\alpha_z} \times \hat{\alpha_z} sin(\theta)] = \hat{\alpha_z} \times \hat{\alpha_r} cos(\theta).$$

Using the determinant definition of the dot product as before,

$$\hat{\alpha_z} \times \hat{\alpha_z} cos(\theta) = \begin{vmatrix} \hat{\alpha_r} & \hat{\alpha_\phi} & \hat{\alpha_z} \\ 0 & 0 & -1 \\ cos(\theta) & 0 & 0 \end{vmatrix} = -\hat{\alpha_\phi} cos(\theta),$$

which also corresponds with the directional component given by the right-hand rule.

2. a) The scalar component of $\vec{B}$ in the direction of $\vec{A}$ is found by means of the dot product between $\vec{B}$ and the unit vector pointing in the direction of $\vec{A}$. First, finding the unit vector pointing in the direction of $\vec{A}$,

$$\hat{\alpha_A} = \frac{\vec{A}}{|\vec{A}|} = \hat{\alpha_x}\frac{4}{\sqrt{21}} + \hat{\alpha_y}\frac{2}{\sqrt{21}} - \hat{\alpha_z}\frac{1}{\sqrt{21}}.$$

Then, the scalar component of $\vec{B}$ in the direction of $\vec{A}$ is

$$\vec{B} \cdot \hat{\alpha_A} = \frac{-8}{\sqrt{21}} - \frac{3}{\sqrt{21}} = -\frac{11}{\sqrt{21}}.$$

b) To find the vector component or projection of $\vec{B}$ in the direction of $\vec{A}$, we must find the vector whose magnitude is the scalar projection of $\vec{B}$ on $\vec{A}$ with the same direction of $\vec{A}$. To find this, we may simply scale the unit vector pointing in the $\vec{A}$ direction by the scalar component of $\vec{B}$ in the $\vec{A}$ direction,

$$(\vec{B} \cdot \hat{\alpha_A})\hat{\alpha_A} = -\hat{\alpha_x}\frac{44}{21} - \hat{\alpha_y}\frac{22}{21} + \hat{\alpha_z}\frac{11}{21}.$$

c) Finally, to find the vector component of $\vec{B}$ in the direction of A, we may use the property that the sum of $\vec{B}$'s vector component parallel to and perpendicular to the direction of A will equal $\vec{B}$. In other words, $\vec{B}$ is made up of a component parallel to the $\vec{A}$ direction and another component orthogonal to the $\vec{A}$ direction. Therefore, the component orthogonal to the $\vec{A}$ direction is given by

$$\vec{B} - [(\vec{B} \cdot \hat{\alpha_A})\hat{\alpha_A}] = \hat{\alpha_x}\frac{2}{21} + \hat{\alpha_y}\frac{22}{21} + \hat{\alpha_z}\frac{52}{21}.$$

3. Given the point $P(1, 2, 3)$ in rectangular coordinates, we may first convert P to cylindrical coordinates of the form $P(r, \phi, z)$ by using the coordinate transformations $r = \sqrt[+]{x^2 + y^2}, \phi = atan2(\frac{y}{x})$, and $z = z$. Therefore,

$$P_{rectangular}(1, 2, 3) = P_{cylindrical}(2.236, 63.43°, 3).$$

Converting to spherical coordinates, we may use the coordinate transformations $R = \sqrt[+]{x^2 + y^2 + z^2}, \theta = atan2(\frac{\sqrt[+]{x^2+y^2}}{z})$, and $\phi = atan2(\frac{y}{x})$. Therefore,

$$P_{rectangular}(1, 2, 3) = P_{cylindrical}(3.7417, 36.70°, 63.43°).$$